

Acids and Bases - Answers

Task 1

Hydrogen chloride	Hydrochloric acid
Sulfur dioxide	Sulfuric acid
Carbon dioxide	Carbonic acid
Nitrogen dioxide	Nitric acid

Task 2

- Sulfur + Oxygen → Sulfur Dioxide
 $\text{S}_{(s)} + \text{O}_{2(g)} \rightarrow \text{SO}_{2(g)}$
- Sulfur Dioxide + Oxygen → Sulfur Trioxide
 $2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{SO}_{3(g)}$
- Sulfur Trioxide + Water → Sulfuric acid
 $\text{SO}_{3(g)} + \text{H}_2\text{O}_{(l)} \rightarrow \text{H}_2\text{SO}_{4(aq)}$

Task 3

- Potassium + Water → Potassium Hydroxide + Hydrogen
 $2\text{K}_{(s)} + 2\text{H}_2\text{O}_{(l)} \rightarrow 2\text{KOH}_{(aq)} + \text{H}_{2(g)}$
- Sodium + Water → Sodium Hydroxide + Hydrogen
 $2\text{Na}_{(s)} + 2\text{H}_2\text{O}_{(l)} \rightarrow 2\text{NaOH}_{(aq)} + \text{H}_{2(g)}$
- Calcium + Water → Calcium Hydroxide + Hydrogen
 $\text{Ca}_{(s)} + 2\text{H}_2\text{O}_{(l)} \rightarrow \text{Ca(OH)}_{2(aq)} + \text{H}_{2(g)}$

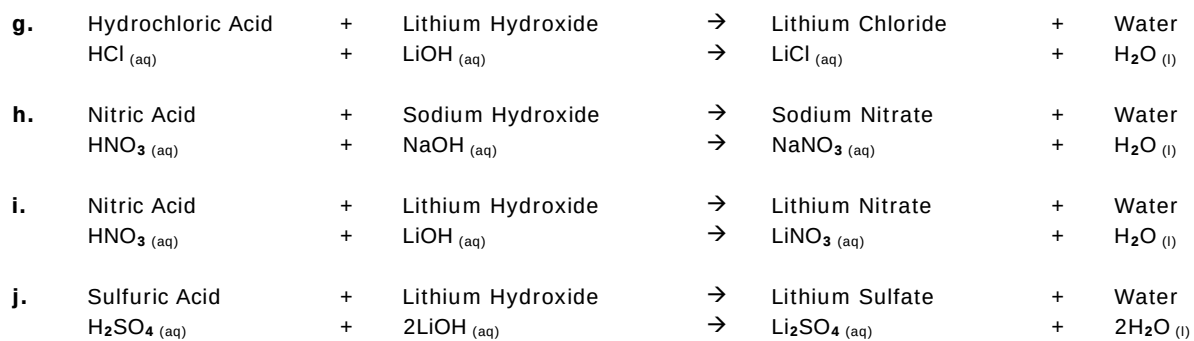
Task 4



Task 5

- Hydrochloric Acid + Sodium Hydroxide → Sodium Chloride + Water
 $\text{HCl}_{(aq)} + \text{NaOH}_{(aq)} \rightarrow \text{NaCl}_{(aq)} + \text{H}_2\text{O}_{(l)}$
- Sulfuric Acid + Potassium Hydroxide → Potassium Sulfate + Water
 $\text{H}_2\text{SO}_{4(aq)} + 2\text{KOH}_{(aq)} \rightarrow \text{K}_2\text{SO}_{4(aq)} + 2\text{H}_2\text{O}_{(l)}$
- Nitric Acid + Calcium Hydroxide → Calcium Nitrate + Water
 $2\text{HNO}_{3(aq)} + \text{Ca(OH)}_{2(aq)} \rightarrow \text{Ca(NO}_3)_2(aq) + 2\text{H}_2\text{O}_{(l)}$
- Hydrochloric Acid + Calcium Hydroxide → Calcium Chloride + Water
 $2\text{HCl}_{(aq)} + \text{Ca(OH)}_{2(aq)} \rightarrow \text{CaCl}_{2(aq)} + 2\text{H}_2\text{O}_{(l)}$
- Nitric Acid + Potassium Hydroxide → Potassium Nitrate + Water
 $\text{HNO}_{3(aq)} + \text{KOH}_{(aq)} \rightarrow \text{KNO}_3(aq) + \text{H}_2\text{O}_{(l)}$
- Sulfuric Acid + Calcium Hydroxide → Calcium Sulfate + Water
 $\text{H}_2\text{SO}_{4(aq)} + \text{Ca(OH)}_{2(aq)} \rightarrow \text{CaSO}_4(aq) + 2\text{H}_2\text{O}_{(l)}$

Task 5 (continued)



Task 6

